Power Series and Interval of Convergence

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Finding the radius of convergence**

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ using the Ratio Test.

$\lim_{n\to\infty} \left|\frac{x^{n+1}/(n+1)}{x^n/n}\right| = \lim_{n\to\infty} \frac{n|x|}{n+1} = |x|$. Converges when $|x| < 1$: radius of convergence $R = 1$.

2 **[Foundational]** • **Target: Finding the radius of convergence**

Find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

$\lim_{n\to\infty} \left|\frac{x^{n+1}/(n+1)!}{x^n/n!}\right| = \lim_{n\to\infty} \frac{|x|}{n+1} = 0 < 1$ for all x. Radius of convergence $R = \infty$ (converges everywhere).

3 **[Foundational]** • **Target: Finding the interval of convergence (testing endpoints)**

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n}$ (radius $R = 1$ found earlier), by testing $x = 1$ and $x = -1$.

At $x = 1$: $\sum \frac{1}{n}$ diverges (harmonic). At $x = -1$: $\sum \frac{(-1)^n}{n}$ converges (alternating harmonic, by AST). Interval of convergence: $[-1, 1)$.

4 **[Foundational]** • **Target: Finding the interval of convergence (testing endpoints)**

Find the interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2}$.

Ratio Test: $\lim \left|\frac{(x-2)^{n+1}/(n+1)^2}{(x-2)^n/n^2}\right| = |x-2| < 1 \Rightarrow 1 < x < 3$. At $x = 1$: $\sum \frac{(-1)^n}{n^2}$ converges absolutely. At $x = 3$: $\sum \frac{1}{n^2}$ converges. Interval: $[1, 3]$.

5 **[Foundational]** • **Target: Representing functions as power series**

Write the power series for $\frac{1}{1-x}$ using the geometric series formula, and state its interval of convergence.

$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$, valid for $|x| < 1$, i.e. interval of convergence $(-1, 1)$.

6 **[Foundational]** • **Target: Representing functions as power series**

Use the geometric series for $\frac{1}{1-x}$ to find a power series for $\frac{1}{1+x^2}$.

Substitute $x \to -x^2$: $\frac{1}{1-(-x^2)} = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n} = 1 - x^2 + x^4 - x^6 + \cdots$.

7 **[Foundational]** • **Target: Differentiating and integrating power series**

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series for $\frac{1}{(1-x)^2}$ by differentiating term by term.

$\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$. Differentiating the series: $\sum_{n=1}^{\infty} n x^{n-1}$. So $\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + \cdots$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Differentiating and integrating power series

Using $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$, find a power series for $\tan^{-1} x$ by integrating term by term.

$\tan^{-1} x = \int \frac{1}{1+x^2}\,dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C$. Since $\tan^{-1} 0 = 0$, $C = 0$: $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$.

9 [Intermediate] • Target: More radius/interval of convergence practice

Find the radius and interval of convergence of $\sum_{n=0}^{\infty} n!\, x^n$.

$\lim_{n\to\infty} \left|\frac{(n+1)!\,x^{n+1}}{n!\,x^n}\right| = \lim_{n\to\infty} (n+1)|x| = \infty$ for any $x \neq 0$. Converges only at $x = 0$: $R = 0$, interval $\{0\}$.

10 [Intermediate] • Target: More radius/interval of convergence practice

Find the radius and interval of convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n (x+1)^n}{n \cdot 3^n}$.

$\lim_{n\to\infty} \left|\frac{(x+1)^{n+1}/[(n+1)3^{n+1}]}{(x+1)^n/(n3^n)}\right| = \frac{|x+1|}{3} < 1 \Rightarrow |x+1| < 3 \Rightarrow -4 < x < 2$; $R = 3$. At $x = 2$: $\sum \frac{(-1)^n}{n}$ converges. At $x = -4$: $\sum \frac{1}{n}$ diverges. Interval: $(-4, 2]$.

11 [Intermediate] • Target: Visualizing a power series approximation

The geometric series $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$ converges on $(-1, 1)$. Compare the graph of $f(x) = \frac{1}{1-x}$ to its degree-3 partial sum $P_3(x) = 1 + x + x^2 + x^3$ on $[-0.8, 0.8]$, and describe where they agree most closely.

The two curves nearly overlap near $x = 0$ and diverge visibly as $x \to \pm 0.8$ (closer to the boundary of the interval of convergence), since the partial sum's approximation quality worsens near the edges of convergence.

12 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

State the theorem describing what happens to the radius of convergence when a power series is differentiated or integrated term by term.

Differentiating or integrating a power series term by term does not change its radius of convergence R (though convergence AT the endpoints $x = \pm R$ may change).

13 [Intermediate] • Target: Term-by-term operations preserve the radius of convergence

Given that $\sum_{n=1}^{\infty} \frac{x^n}{n}$ has radius of convergence $R = 1$, state the radius of convergence of its derivative $\sum_{n=1}^{\infty} x^{n-1}$ without recomputing.

By the term-by-term differentiation theorem, the radius of convergence is unchanged: $R = 1$.

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14 [Intermediate] • Target: Building a new series by combining known ones

Using $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ and $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, find the power series for $\frac{2}{1-x^2}$ by adding the two series.

$\frac{1}{1-x} + \frac{1}{1+x} = \frac{(1+x)+(1-x)}{1-x^2} = \frac{2}{1-x^2}$. Adding the series: $\sum_{n=0}^{\infty} x^n + \sum_{n=0}^{\infty} (-1)^n x^n = \sum_{n=0}^{\infty} [1+(-1)^n] x^n$, which is $2x^n$ for even n and 0 for odd n: $2 + 2x^2 + 2x^4 + \cdots$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Ratio Test with factorial and power combinations • Cross-Lesson: 3.01 Parametric Equations

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{2^n x^n}{n^2}$.

$\lim_{n\to\infty} \left| \frac{2^{n+1}x^{n+1}/(n+1)^2}{2^n x^n/n^2} \right| = \lim_{n\to\infty} \frac{2n^2|x|}{(n+1)^2} = 2|x| < 1 \Rightarrow |x| < \frac{1}{2}$. $R = \frac{1}{2}$.

16 [Challenging] • Target: Ratio Test with factorial and power combinations • Cross-Lesson: 3.02 Vector-Valued Functions

For the series above, test both endpoints $x = \frac{1}{2}$ and $x = -\frac{1}{2}$ to find the full interval of convergence.

At $x = \frac{1}{2}$: $\sum \frac{2^n (1/2)^n}{n^2} = \sum \frac{1}{n^2}$, converges (p-series, $p = 2$). At $x = -\frac{1}{2}$: $\sum \frac{(-1)^n}{n^2}$, converges absolutely. Interval: $\left[-\frac{1}{2}, \frac{1}{2}\right]$.

17 [Challenging] • Target: Using a power series to approximate a definite integral • Cross-Lesson: 4.01 Polar Curves and Derivatives

Using the power series $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$ found earlier, estimate $\tan^{-1}(0.5)$ using the first three terms.

$\tan^{-1}(0.5) \approx 0.5 - \frac{(0.5)^3}{3} + \frac{(0.5)^5}{5} = 0.5 - 0.04167 + 0.00625 \approx 0.4646$ (true value ≈ 0.4636, a close match).

18 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 4.02 Area in Polar Coordinates

Evaluate $\sum_{n=0}^{\infty} 3(\frac{1}{4})^n$.

Method · Geometric Series Sum Evaluation: Step 1 · Identify series parameters: Initial term ($n = 0$): $a = 3 \cdot \left(\frac{1}{4}\right)^0 = 3 \cdot 1 = 3$. Common ratio: $r = \frac{1}{4}$. Step 2 · Check convergence condition: $|r| = \frac{1}{4} < 1$, so the series converges absolutely. Step 3 · Compute the sum: $S = \frac{a}{1-r} = \frac{3}{1-1/4} = \frac{3}{3/4} = 3 \cdot \frac{4}{3} = \mathbf{4}$.

19 [Challenging] • Target: More radius/interval of convergence practice • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Evaluate $\int \frac{1}{1+x^2}\,dx$.

Theorem (Standard Inverse Trigonometric Antiderivative): By the Fundamental Theorem of Calculus, integration is the reverse operation of differentiation. Since $\frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$, we have: $\int \frac{1}{1+x^2}\,dx = \arctan\mathbf{(x) + C}$. General Form: More broadly, for any nonzero real constant a: $\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$.

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20 [Challenging] • Target: Visualizing a power series approximation • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Does $\sum_{n=0}^{\infty} (0.9)^n$ converge, and to what?

Method · Geometric Series Convergence & Sum: Step 1 · Identify series parameters: Initial term: $a = (0.9)^0 = 1$. Common ratio: $r = 0.9 = \frac{9}{10}$. Step 2 · Test convergence: $|r| = 0.9 < 1$, therefore the series converges. Step 3 · Calculate exact sum: $S = \frac{a}{1-r} = \frac{1}{1-0.9} = \frac{1}{0.1} = \mathbf{10}$. Conclusion: Converges to **10**.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Finding the radius of convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{6^n}$.

a. 12 **b.** $\frac{1}{6}$

c. 6 ✓ **d.** 1

$\lim_{n\to\infty} \left|\frac{x^{n+1}/6^{n+1}}{x^n/6^n}\right| = \frac{|x|}{6} < 1 \Rightarrow |x| < 6.\ R = 6.$

22 [Foundational] • Target: Finding the interval of convergence (testing endpoints)

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{18^n}$.

a. 1 **b.** 18 ✓

c. 36 **d.** $\frac{1}{18}$

$\lim_{n\to\infty} \left|\frac{x^{n+1}/18^{n+1}}{x^n/18^n}\right| = \frac{|x|}{18} < 1 \Rightarrow |x| < 18.\ R = 18.$

23 [Foundational] • Target: Representing functions as power series

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{5^n}$.

a. 1 **b.** 10

c. 5 ✓ **d.** $\frac{1}{5}$

$\lim_{n\to\infty} \left|\frac{x^{n+1}/5^{n+1}}{x^n/5^n}\right| = \frac{|x|}{5} < 1 \Rightarrow |x| < 5.\ R = 5.$

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24 [Foundational] • Target: Differentiating and integrating power series

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{20^n}$.

a. 1

b. 20 ✓

c. 40

d. $\frac{1}{20}$

$\lim_{n\to\infty} \left|\frac{x^{n+1}/20^{n+1}}{x^n/20^n}\right| = \frac{|x|}{20} < 1 \Rightarrow |x| < 20.\ R = 20.$

25 [Foundational] • Target: More radius/interval of convergence practice

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{14^n}$.

a. 28

b. 14 ✓

c. $\frac{1}{14}$

d. 1

$\lim_{n\to\infty} \left|\frac{x^{n+1}/14^{n+1}}{x^n/14^n}\right| = \frac{|x|}{14} < 1 \Rightarrow |x| < 14.\ R = 14.$

26 [Foundational] • Target: Visualizing a power series approximation

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{4^n}$.

a. 4 ✓

b. $\frac{1}{4}$

c. 1

d. 8

$\lim_{n\to\infty} \left|\frac{x^{n+1}/4^{n+1}}{x^n/4^n}\right| = \frac{|x|}{4} < 1 \Rightarrow |x| < 4.\ R = 4.$

27 [Foundational] • Target: Term-by-term operations preserve the radius of convergence

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{15^n}$.

a. 1

b. $\frac{1}{15}$

c. 15 ✓

d. 30

$\lim_{n\to\infty} \left|\frac{x^{n+1}/15^{n+1}}{x^n/15^n}\right| = \frac{|x|}{15} < 1 \Rightarrow |x| < 15.\ R = 15.$

28 [Foundational] • Target: Building a new series by combining known ones

Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+6)^n$.

a. $\frac{1}{7}$ ✓

b. 7

c. $\frac{8}{7}$

d. $\frac{1}{8}$

$\lim_{n\to\infty} \left|\frac{7^{n+1}(x+6)^{n+1}}{7^n(x+6)^n}\right| = 7|x+6| < 1 \Rightarrow |x+6| < \frac{1}{7}.\ R = \frac{1}{7}.$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Ratio Test with factorial and power combinations**

Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+3)^n$.

a. 7 **b.** $\frac{8}{7}$

c. $\frac{1}{7}$ ✓ **d.** $\frac{1}{8}$

$\lim_{n\to\infty}\left|\frac{7^{n+1}(x+3)^{n+1}}{7^n(x+3)^n}\right| = 7|x+3| < 1 \Rightarrow |x+3| < \frac{1}{7}$. $R = \frac{1}{7}$.

30 **[Intermediate]** • **Target: Using a power series to approximate a definite integral**

Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+7)^n$.

a. $\frac{1}{3}$ **b.** $\frac{3}{2}$

c. $\frac{1}{2}$ ✓ **d.** 2

$\lim_{n\to\infty}\left|\frac{2^{n+1}(x+7)^{n+1}}{2^n(x+7)^n}\right| = 2|x+7| < 1 \Rightarrow |x+7| < \frac{1}{2}$. $R = \frac{1}{2}$.

31 **[Intermediate]** • **Target: Finding the Radius of Convergence**

Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+4)^n$.

a. $\frac{1}{2}$ ✓ **b.** 2

c. $\frac{3}{2}$ **d.** $\frac{1}{3}$

$\lim_{n\to\infty}\left|\frac{2^{n+1}(x+4)^{n+1}}{2^n(x+4)^n}\right| = 2|x+4| < 1 \Rightarrow |x+4| < \frac{1}{2}$. $R = \frac{1}{2}$.

32 **[Intermediate]** • **Target: Testing Interval Endpoints**

Find the radius of convergence of $\sum_{n=1}^{\infty} 2^n(x+8)^n$.

a. $\frac{1}{2}$ ✓ **b.** $\frac{3}{2}$

c. $\frac{1}{3}$ **d.** 2

$\lim_{n\to\infty}\left|\frac{2^{n+1}(x+8)^{n+1}}{2^n(x+8)^n}\right| = 2|x+8| < 1 \Rightarrow |x+8| < \frac{1}{2}$. $R = \frac{1}{2}$.

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33 [Intermediate] • Target: Term-by-Term Differentiation of Power Series

Find the radius of convergence of $\sum_{n=1}^{\infty} 7^n(x+8)^n$.

a. $\frac{8}{7}$

b. $\frac{1}{7}$ ✓

c. 7

d. $\frac{1}{8}$

$\lim_{n\to\infty}\left|\frac{7^{n+1}(x+8)^{n+1}}{7^n(x+8)^n}\right| = 7|x+8| < 1 \Rightarrow |x+8| < \frac{1}{7}$. $R = \frac{1}{7}$.

34 [Intermediate] • Target: Term-by-Term Integration of Power Series

Find the radius of convergence of $\sum_{n=1}^{\infty} 6^n(x+3)^n$.

a. $\frac{1}{6}$ ✓

b. 6

c. $\frac{1}{7}$

d. $\frac{7}{6}$

$\lim_{n\to\infty}\left|\frac{6^{n+1}(x+3)^{n+1}}{6^n(x+3)^n}\right| = 6|x+3| < 1 \Rightarrow |x+3| < \frac{1}{6}$. $R = \frac{1}{6}$.

35 [Intermediate] • Target: Finding the radius of convergence

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-1)^n}{4^n n}$.

a. **[-3,5)** ✓

b. (-3,5]

c. (-3,5)

d. [-3,5]

Ratio Test gives $|x-1| < 4$, so $-3 < x < 5$ ($R = 4$). Testing endpoints: at $x = 5$: $\sum \frac{1}{n}$ diverges; at $x = -3$: $\sum \frac{(-1)^n}{n}$ converges by the AST. Interval: $[-3, 5)$.

36 [Intermediate] • Target: Finding the interval of convergence (testing endpoints)

Find the full interval of convergence of $\sum_{n=1}^{\infty} \frac{(x-5)^n}{4^n n}$.

a. (1,9]

b. **[1,9)** ✓

c. [1,9]

d. (1,9)

Ratio Test gives $|x-5| < 4$, so $1 < x < 9$ ($R = 4$). Testing endpoints: at $x = 9$: $\sum \frac{1}{n}$ diverges; at $x = 1$: $\sum \frac{(-1)^n}{n}$ converges by the AST. Interval: $[1, 9)$.

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37 [Intermediate] • Target: Representing functions as power series

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 8^n x^n}{16^n}$.

a. 2 ✓

b. 3

c. $\frac{1}{2}$

d. 128

The polynomial factor n^2 does not affect the radius. $\lim_{n\to\infty} \left|\frac{8^{n+1}x^{n+1}/16^{n+1}}{8^n x^n/16^n}\right| = \frac{8}{16}|x| < 1 \Rightarrow |x| < \frac{16}{8}$. $R = 2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Differentiating and integrating power series • Cross-Lesson: 4.02 Area in Polar Coordinates

Find the radius of convergence of $\sum_{n=1}^{\infty} \frac{n^2 \cdot 6^n x^n}{11^n}$.

a. $\frac{6}{11}$

b. $\frac{17}{6}$

c. $\frac{11}{6}$ ✓

d. 66

The polynomial factor n^2 does not affect the radius. $\lim_{n\to\infty} \left|\frac{6^{n+1}x^{n+1}/11^{n+1}}{6^n x^n/11^n}\right| = \frac{6}{11}|x| < 1 \Rightarrow |x| < \frac{11}{6}$. $R = \frac{11}{6}$.

39 [Challenging] • Target: More radius/interval of convergence practice • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Using $\frac{1}{1-3x} = \sum_{n=0}^{\infty} 3^n x^n$, find the coefficient of x^5.

a. 81

b. 729

c. 15

d. 243 ✓

The coefficient of x^n is 3^n, so the coefficient of x^5 is $3^5 = 243$.

40 [Challenging] • Target: Visualizing a power series approximation • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Using $\frac{1}{1-5x} = \sum_{n=0}^{\infty} 5^n x^n$, find the coefficient of x^3.

a. 25

b. 125 ✓

c. 15

d. 625

The coefficient of x^n is 5^n, so the coefficient of x^3 is $5^3 = 125$.

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41 [Challenging] • Target: Finding the Radius of Convergence • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the radius of convergence R for the power series $\sum_{n=0}^{\infty} \frac{(x-3)^n}{4^n}$.

a. $R = 4$ ✓

b. $R = 1$

c. $R = 3$

d. $R = \infty$

Ratio test: $\lim_{n\to\infty} \left|\frac{(x-3)^{n+1}}{4^{n+1}} \frac{4^n}{(x-3)^n}\right| = \frac{|x-3|}{4} < 1 \Rightarrow |x-3| < 4$. The radius is $R = 4$.

42 [Challenging] • Target: Finding the Radius of Convergence • Cross-Lesson: 1.01 Integration by Parts

Find the radius of convergence of $\sum_{n=1}^{\infty} n!\, x^n$.

a. $R = 0$ **(converges only at** $x = 0$**)** ✓

b. $R = 1$

c. $R = \infty$

d. $R = \frac{1}{e}$

Ratio test: $\lim_{n\to\infty} (n+1)|x| = \infty$ for any $x \neq 0$. The series converges only at its center $x = 0$, so $R = 0$.

43 [Challenging] • Target: Testing Interval Endpoints • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Why must endpoints always be tested individually when finding the interval of convergence?

a. **The Ratio Test yields** $L = 1$ **at the boundary points** $|x - c| = R$ **and is inconclusive** ✓

b. Endpoints always diverge

c. Endpoints always converge

d. To find the center of the series

Because the Ratio Test limit equals 1 at $x = c \pm R$, it yields no information at endpoints, necessitating individual convergence tests.

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44 [Challenging] • Target: Term-by-Term Differentiation of Power Series • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Starting from $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series representation for $\frac{1}{(1-x)^2}$.

a. $\sum_{n=1}^{\infty} nx^{n-1}$ ✓

b. $\sum_{n=0}^{\infty} x^{2n}$

c. $\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$

d. $\sum_{n=1}^{\infty} n^2 x^n$

Differentiate both sides: $\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$. Differentiating the series term-by-term gives $\sum_{n=1}^{\infty} nx^{n-1}$.

45 [Challenging] • Target: Term-by-Term Integration of Power Series • Cross-Lesson: 3.01 Parametric Equations

Integrating $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$ term-by-term yields the Maclaurin series for which function?

a. $\ln(1+x)$ ✓

b. $\arctan x$

c. e^{-x}

d. $\frac{1}{(1+x)^2}$

$\int_0^x \frac{1}{1+t} dt = \ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$

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